Marking Scheme – PA3

110 MARKS Undergraduate

130 MARKS Graduate

Q1 – MNIST Download (2 Marks)

- 1 Mark: Dataset downloaded with torchvision
- 1 Mark: Transforms applied C dataset size printed

Q2 – Splits (2 Marks)

- 1 Mark: Train/Val/Test splits are correct (50k/10k/10k)
- 1 Mark: DataLoader sizes printed

Q3 – Visualisation (2 Marks)

- 1 Mark: visualise_samples implemented with correct arguments
- 1 Mark: At least 2 samples displayed

Q4 – Forward Pass in NumPy (17 Marks)

1. **He Initialization** – 5 Marks
   - 3 Marks: Correct formula
   - 2 Marks: Bias set to zeros
2. **Weight Initialization** – 2 Marks (Correct dimensions per layer)
3. **Activations** – 5 Marks
   - 2 Marks: ReLU correct
   - 3 Marks: Softmax correct (with stability)
4. **Forward Pass** – 5 Marks
   - 3 Marks: Correct probability outputs
   - 2 Marks: Predicted class printed

Q5 – PyTorch Neural Networks (62 Marks)

1. **Count Parameters** – 2 Marks (Binary)
2. **FCNN (>200k params)** – 10 Marks
   - 4 Marks: Correct dimensions / architecture
   - 2 Marks: ReLU activation used
   - 2 Marks: Param count >200k shown
   - 2 Marks: No convolutions used

3. **Hyperparameters** - 5 Marks
    - 2 Marks: Correct loss function
    - 2 Marks: Optimizer + LR set
    - 1 Mark: Explanation in markdown
4. **Training s Evaluation Functions** - 20 Marks
    - 7.5 Marks: train function implemented
    - 7.5 Marks: evaluate function implemented
    - 5 Marks: **Average loss accuracy correctly computed**
    - -2 Marks: Per wrong explanation of zero_grad(), backward(), step()
5. **Training Loop** - 15 Marks
    - 5 Marks: Training/Validation loss C accuracy printed for each epoch
    - 5 Marks: Best model saved properly
    - 5 Marks: Curves plotted and labelled
6. **Evaluation on Test Set** - 10 Marks
    - 3 Marks: Accuracy ≥95%
    - 3 Marks: F1 C Recall ≥95%
    - 2 Marks: Classification report + confusion matrix
    - 2 Marks: ROC curve + incorrect samples visualised
7. **OOD Evaluation** - 15 Marks
    - 13 Marks: Image preprocessing + visualization + Forward Pass
    - 1 Marks: Predictions shown
    - 1 Mark: Comment on performance

Q6 - Data Pruning s Smaller Model (25 Marks)

1. **Smaller FCNN (<110k params)** - 5 Marks
    - 3 Marks: Correct architecture
    - 2 Marks: Param count printed
2. Skip
3. **Hyperparameters** - 2 Marks (Binary)
4. **Bottom-k Dataloader** - 5 Marks (Subset used correctly)

5. **Training Loop** – 5 Marks (Metrics printed, works end-to-end)
6. **Evaluation** – 8 Marks
    - 3.5 Marks: Accuracy ≥95%
    - 3.5 Marks: F1 C Recall ≥95%
    - 1 Marks: Confusion matrix + performance comparison

- **-5 if average loss not computed correctly**
- **-5 if accuracy computed incorrectly**

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Graduate Question MS:

Per Activation Function 1 Mark -	**3 Marks (Binary)**
Backward Pass - 17 Marks:	
Correct Derivative of Loss Function - BCE -	**2 Marks**
Correct Shapes of dW, dB, L, m -	**4 Marks (1 each)**
Correct For Loop for Backpropogation:	
Goes from L to 1 -	**-5 if incorrect**
Correctness of dW	**3 Marks**
Correctness of dB	**3 Marks**
Correctness of dZ (for next iteration) -	**5 Marks**